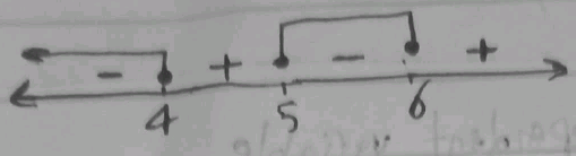
Group 3 (20) 20241700556 عمر حسن رجاء الفطايري

1) a) $\frac{(x-4)(x-5)}{x-6} \leq 0$



$x \in (-\infty, 4] \cup [5, 6)$ #

b) if $\frac{x}{2} - 5 \geq 0 \Rightarrow x \geq 10$

$2 \leq \frac{x}{2} - 5 \leq 6$

$4 \leq x - 10 \leq 12$

$14 \leq x \leq 22$

$x \in (14, 22) \cap [10, \infty) =$ [illegible] $(14, 22$

if $\frac{x}{2} - 5 \leq 0 \quad x \leq 10$

$2 \leq -(\frac{x}{2} - 5) \leq 6$

$-2 \geq \frac{x}{2} - 5 \geq 6$

$-4 \geq x - 10 \geq -12$

$6 \geq x \geq -2$

$x \in (-2, 6) \cap (-\infty, 10) = (-2, 6]$

$x \in (-2, 6) \cup (14, 22)$ #

c) $2x < 4$

$x < 2$

$x \in (-\infty, 2)$

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2) 1- ~~[illegible]~~ independent variable

2- domain

3- dependent variable

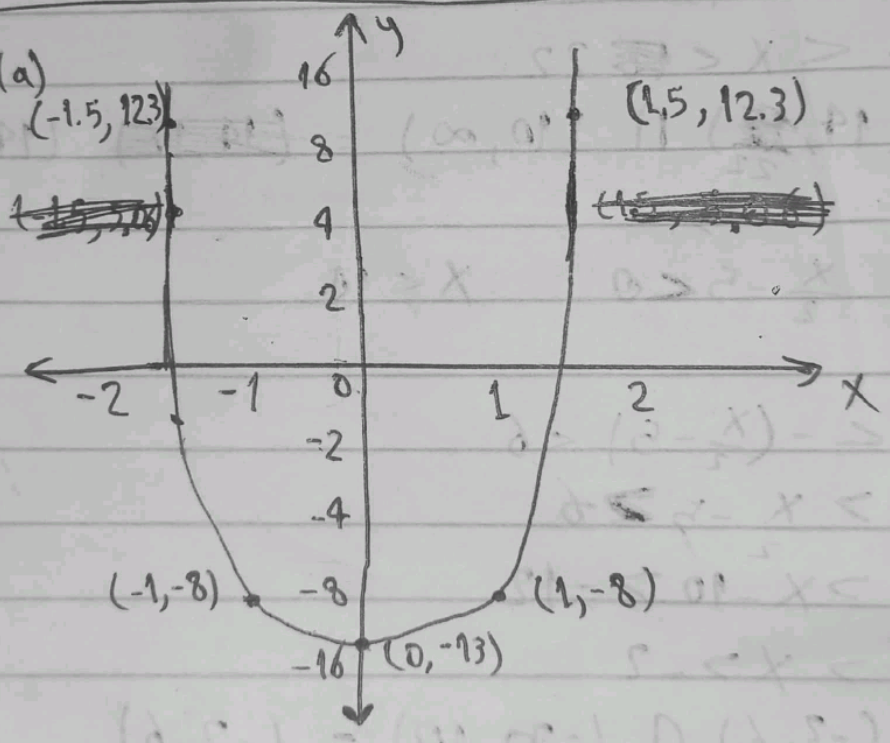
4- range

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3) 1- domain 2- range

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4) (a)

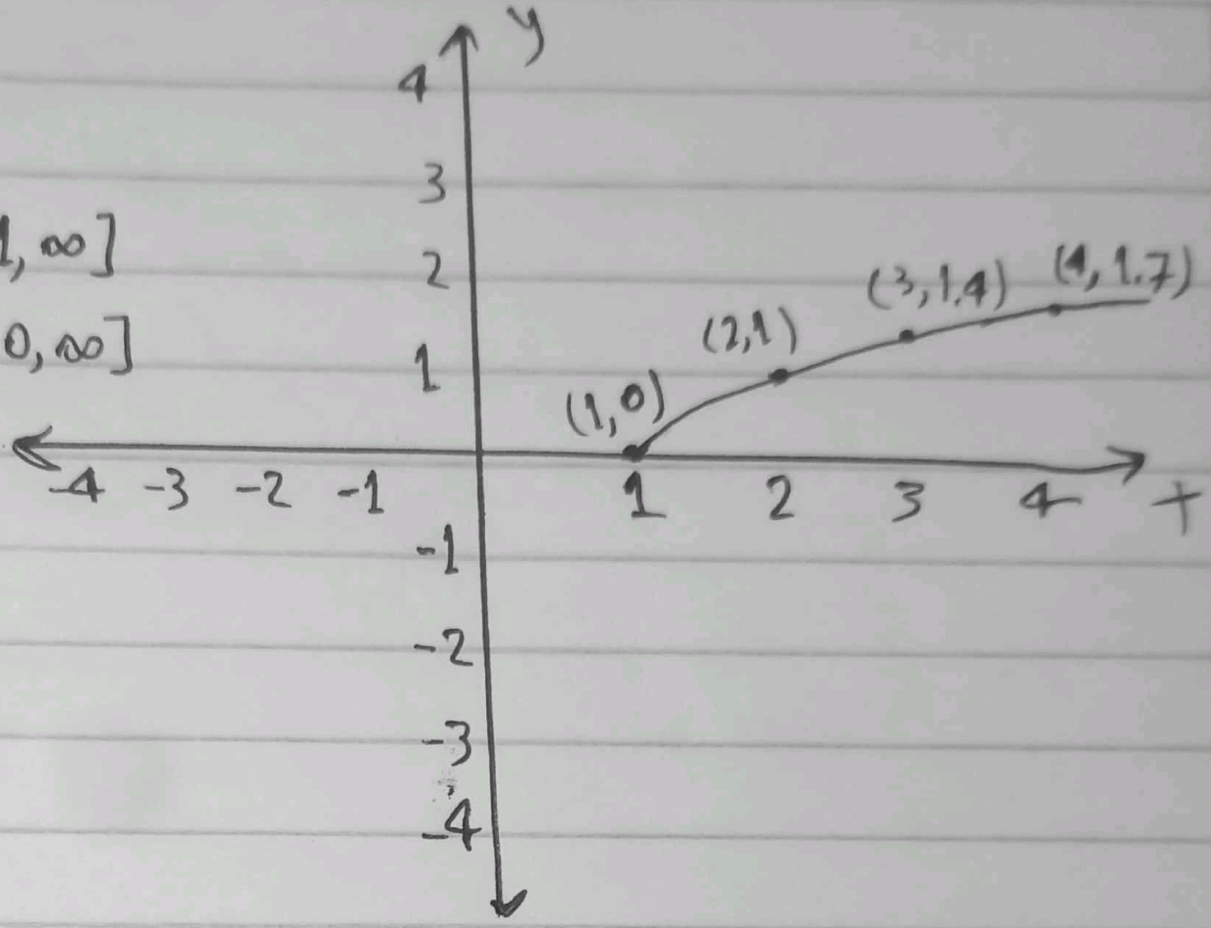


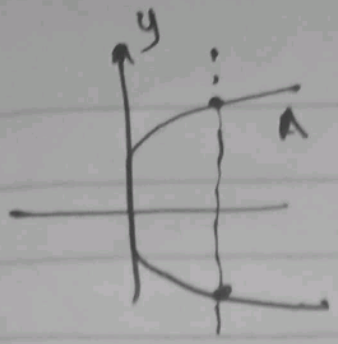
Domain R Range $[-13, \infty)$

4) (b)

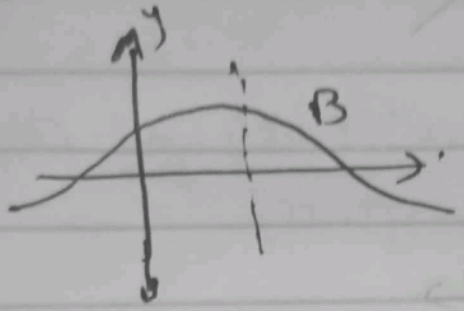
Domain is $[1, \infty]$

Range is $[0, \infty]$



5) 

not a function



a function

B #

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6) $f(g(x)) = \frac{1}{x^3 - 4}$ #

$x^3 \neq 4$

$x \neq \sqrt[3]{4}$

domain $= R - \{\sqrt[3]{4}\}$ #

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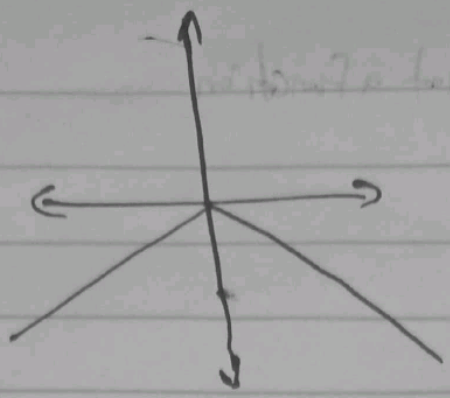
7) $(g \circ f)(x) = \frac{2}{\frac{x}{3x+2}} = \frac{6x+4}{x} = 6 + \frac{4}{x}$

domain = domain of $f(x) \cap$ domain of $(g \circ f)(x)$

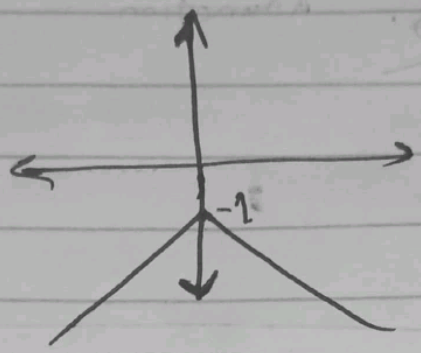
$= R - \{-\frac{2}{3}\} \cap R - \{0\}$

$= R - \{-\frac{2}{3}, 0\}$ #

8) $-f(x)$



$-f(x)-1$



---

9) 1- $R - \{\frac{\pi}{2} + n\pi\}$ where $n \in \mathbb{Z}$

2- R

3- π

4- R

5- $[-1, 1]$

6- 2π

---

10) a) $X > a$

b) $\lim_{x \to a} f(x) = L$

11) a) 2 b) 1 c) 1
d) 1 e) 4 f) doesn't exist because $\lim_{x\to 1^+} g(x) = 5 \neq \lim_{x\to 1^-} g(x) = 3$
g) 4 h) 5 i) 5

12) a) $\lim_{x\to \frac{1}{2}} \frac{(2x-1)(x+3)}{(2x-1)(3x-2)} = \lim_{x\to \frac{1}{2}} \frac{x+3}{3x-2} = \frac{3.5}{-0.5} = -7$ #

b) $\lim_{x\to -3} \frac{(x-2)(x+3)}{(x+3)} = \lim_{x\to -3} x-2 = -5$ #

c) $\lim_{x\to b} \frac{(x-b)((x-b)^{19}-1)}{(x-b)} = \lim_{x\to b} ((x-b)^{19}-1) = -1$ #

d) $\lim_{x\to -6^+} \frac{x+6}{|x+6|} = \frac{+0}{+0} = 1$

$\lim_{x\to -6^-} \frac{x+6}{|x+6|} = \frac{-0}{+0} = -1$

$\lim_{x\to -6} \frac{x+6}{|x+6|}$ doesn't exist

13) a) $\frac{-ve}{+0} = -\infty$

b) $\frac{-ve}{+0} = -\infty$

c) $-\infty$

---

a) $\frac{-ve}{+0} = -\infty$

b) $\frac{-ve}{+0} = -\infty$

c) $-\infty$

---

14) $f(x) = \frac{x(x-6)(x-3)}{x(x-6)} = x-3$

no vertical asymptotes

---

15) (1) $f(x) = \frac{(x-3)(x-1)}{x-1} = x-3$

$\lim_{x \to \infty} f(x) = \infty$ $\lim_{x \to -\infty} f(x) = -\infty$

no horizontal asymptotes

(2) $f(x) = \dfrac{2x(x+2)(x+3)}{x^2(x+2)} = \dfrac{2(x+3)}{x} = 2 + \dfrac{6}{x}$

$\lim_{x \to \infty} f(x) = 2 \qquad \lim_{x \to -\infty} f(x) = 2$

horizontal asymptote at $y = 2$ #

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16) (a) ~~f(x) =~~ $f(-5)$ is undefined $\Rightarrow$ not continuous

(b) $f(1)$ is undefined $\Rightarrow$ not continuous